

Introduction to Musculoskeletal Anatomy: Part 1- introduction and bones

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What makes up the MSK system?

- Bone
- Cartilage
- Muscle
- Connective Tissue
 - Ligaments
 - Fascia



Session Objectives

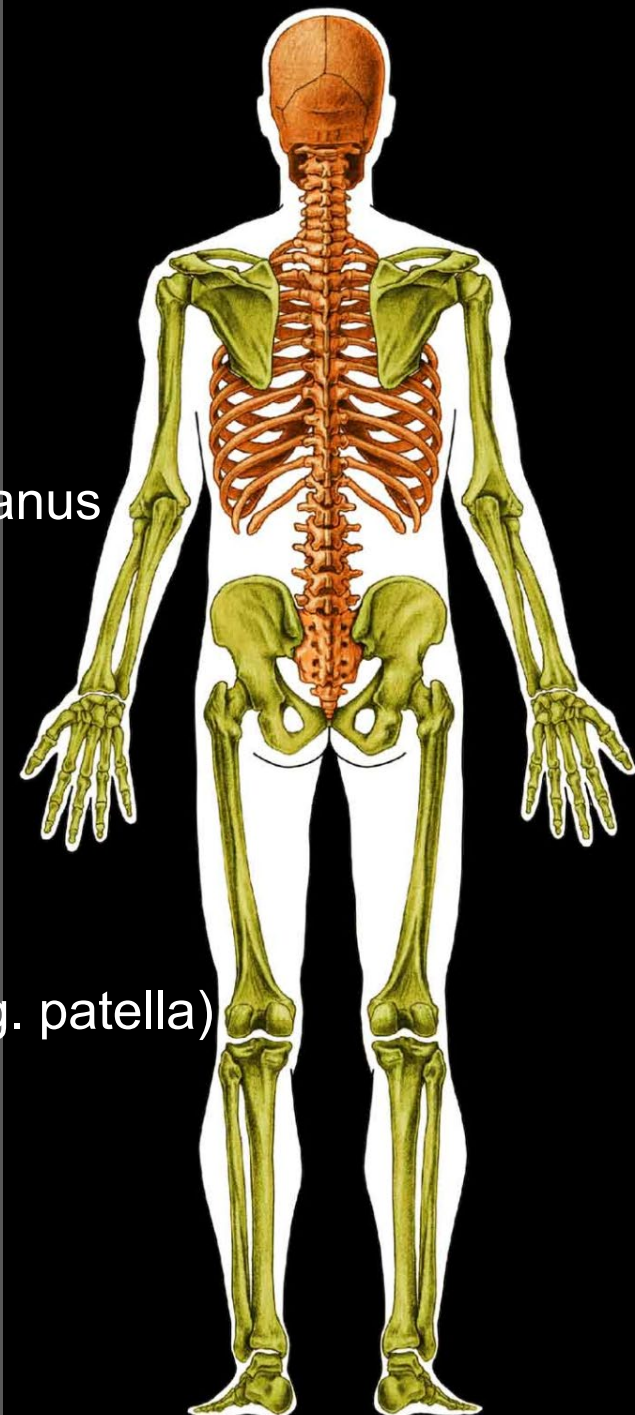
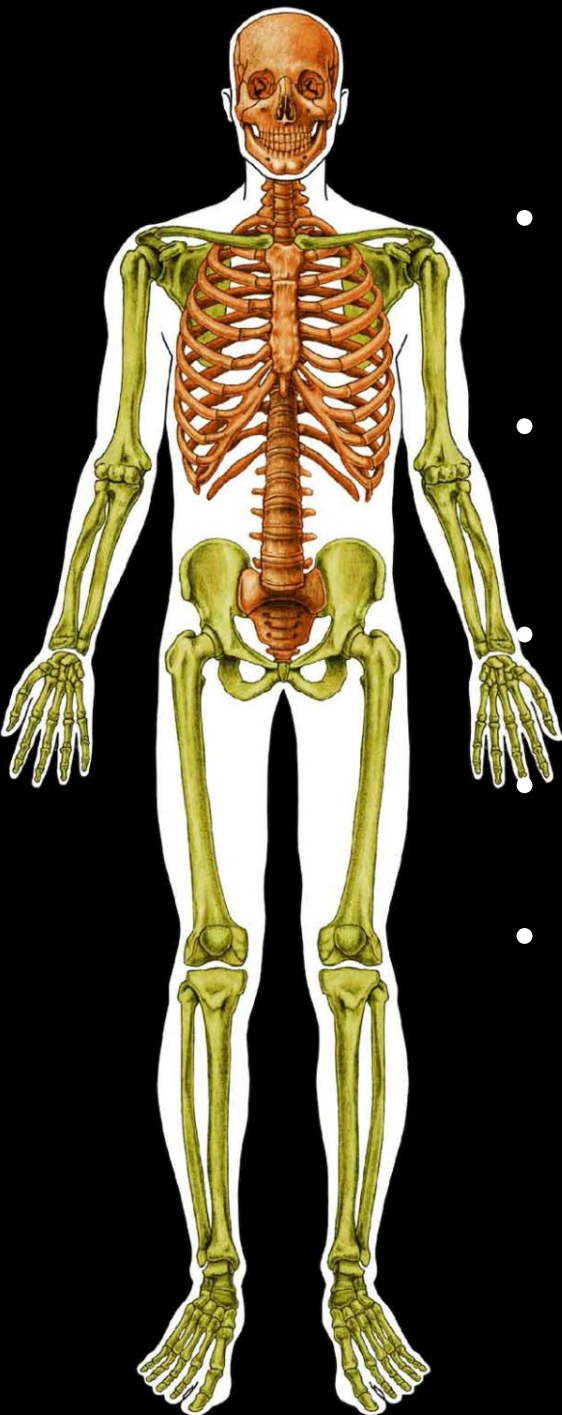
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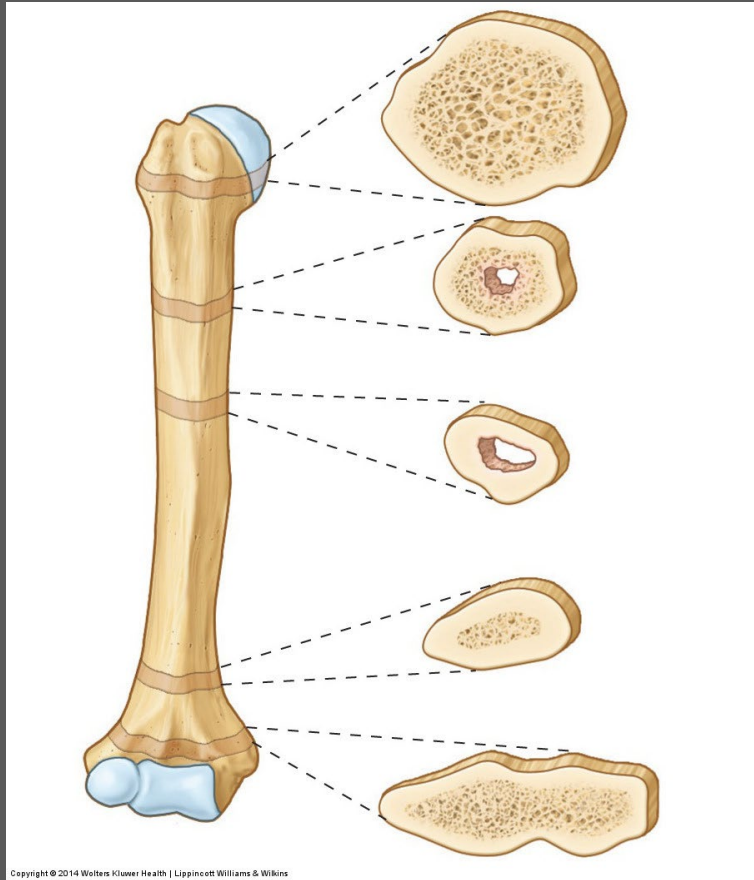
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- Explain the growth, development and morphology of skeletal tissues.
- Describe the types of synovial joints, their movements and distribution.
- Describe the different types of cartilage and their distribution throughout the body.
- Understand the different types of muscle contractions and how muscles contribute to motion (and stasis) in the body.

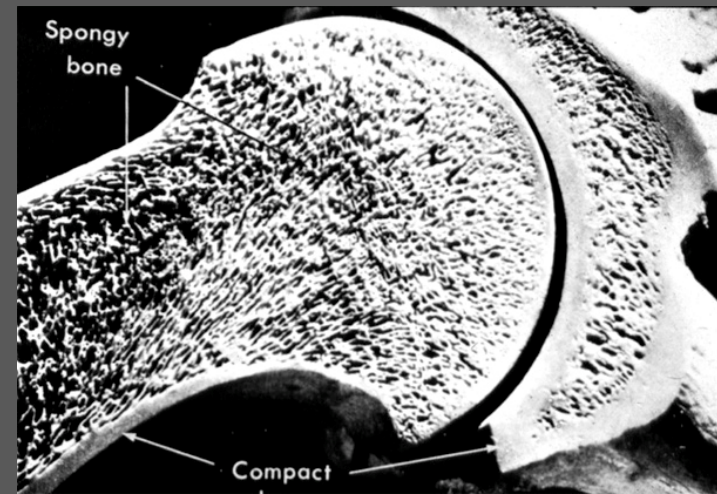
Bones

- Long bones
 - diaphysis (tubular shaft), epiphyses (ends)
- Short bones
 - small blocky; found in manus (hand) and pes (foot)
- Flat bones
 - skull bones, scapula
- Irregular bones
 - vertebrae, pelvic bones
- Sesamoid bones
 - embedded in tendon (e.g. patella)



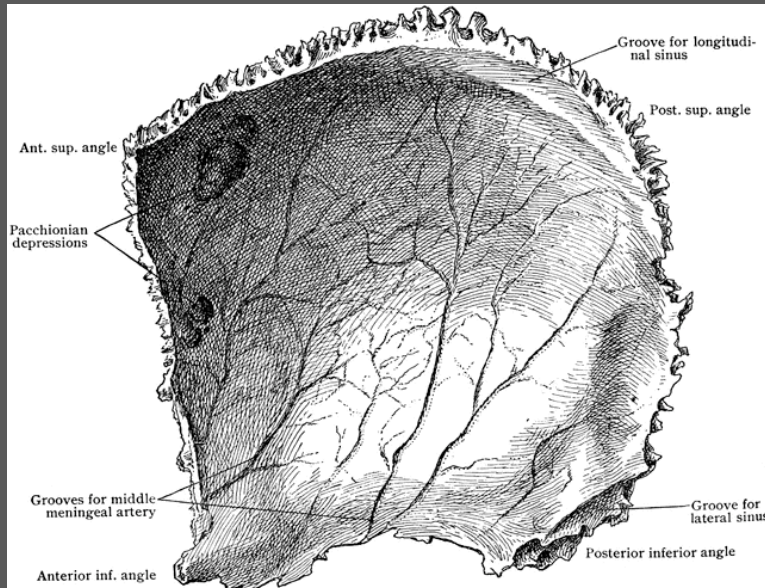


- Compact bone:
 - Superficial layer of dense bone
- Spongy bone:
 - Forms a central mass



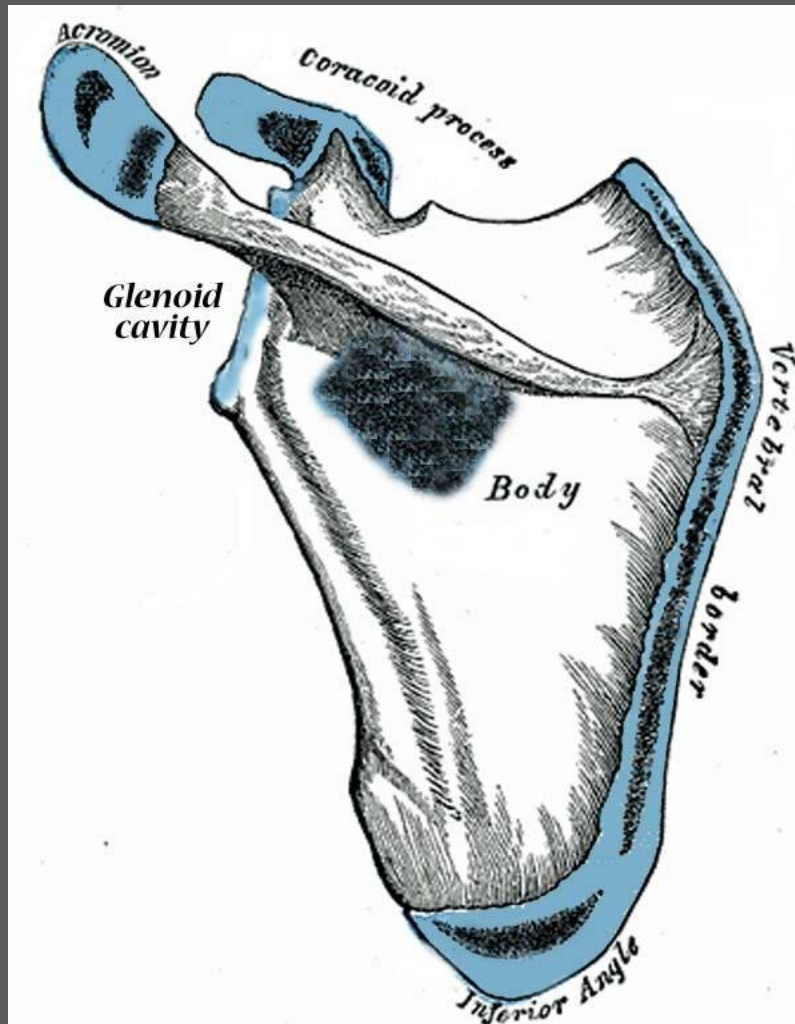
Bones are derived from mesenchyme by two processes

Intramembranous: ossifies directly from mesenchyme;
e.g. flat bones



Endochondral: replaces cartilage derived from mesenchyme
e.g. long bones and short bones

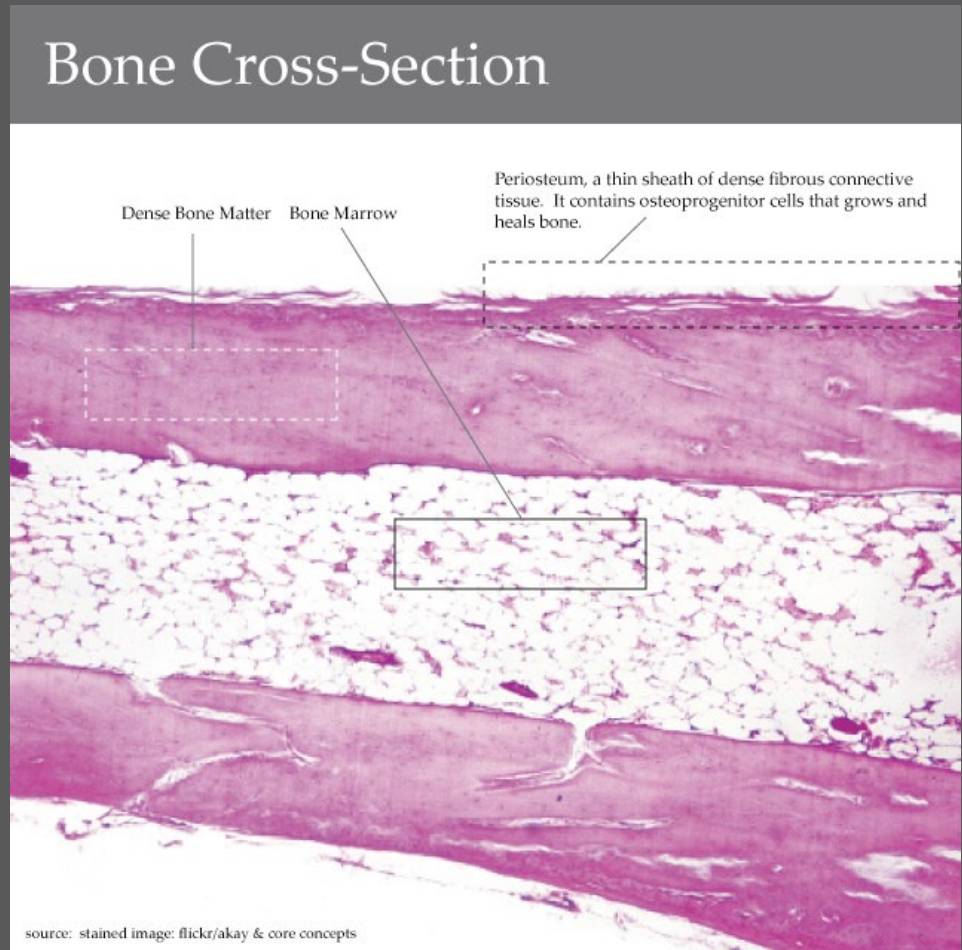




The scapula has endochondral and intramembranous components.

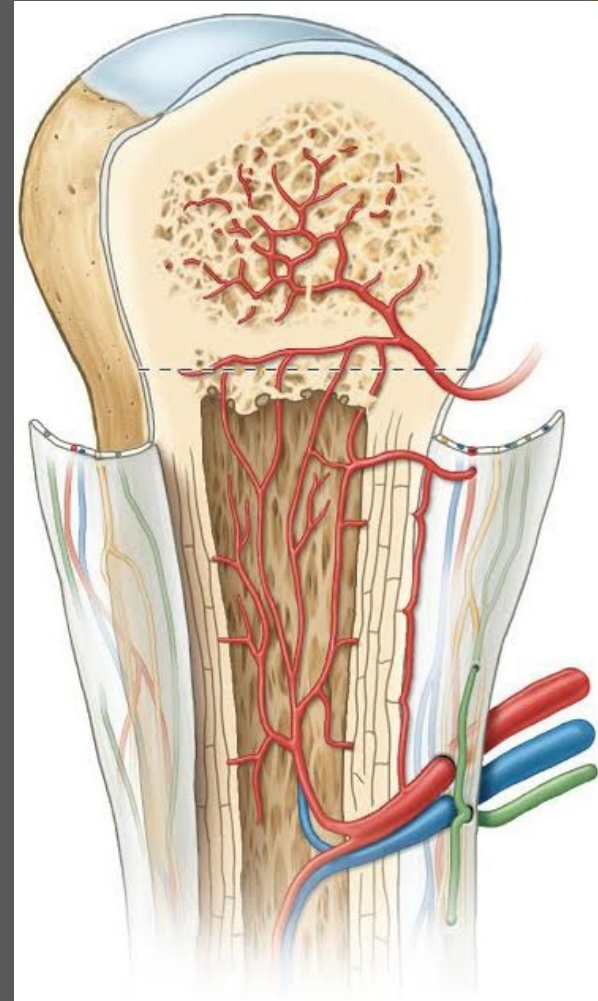
Periosteum

- Fibrous connective tissue surrounding bone that provides...
 - Nourishment
 - Regeneration
 - Attachment
 - Sensation



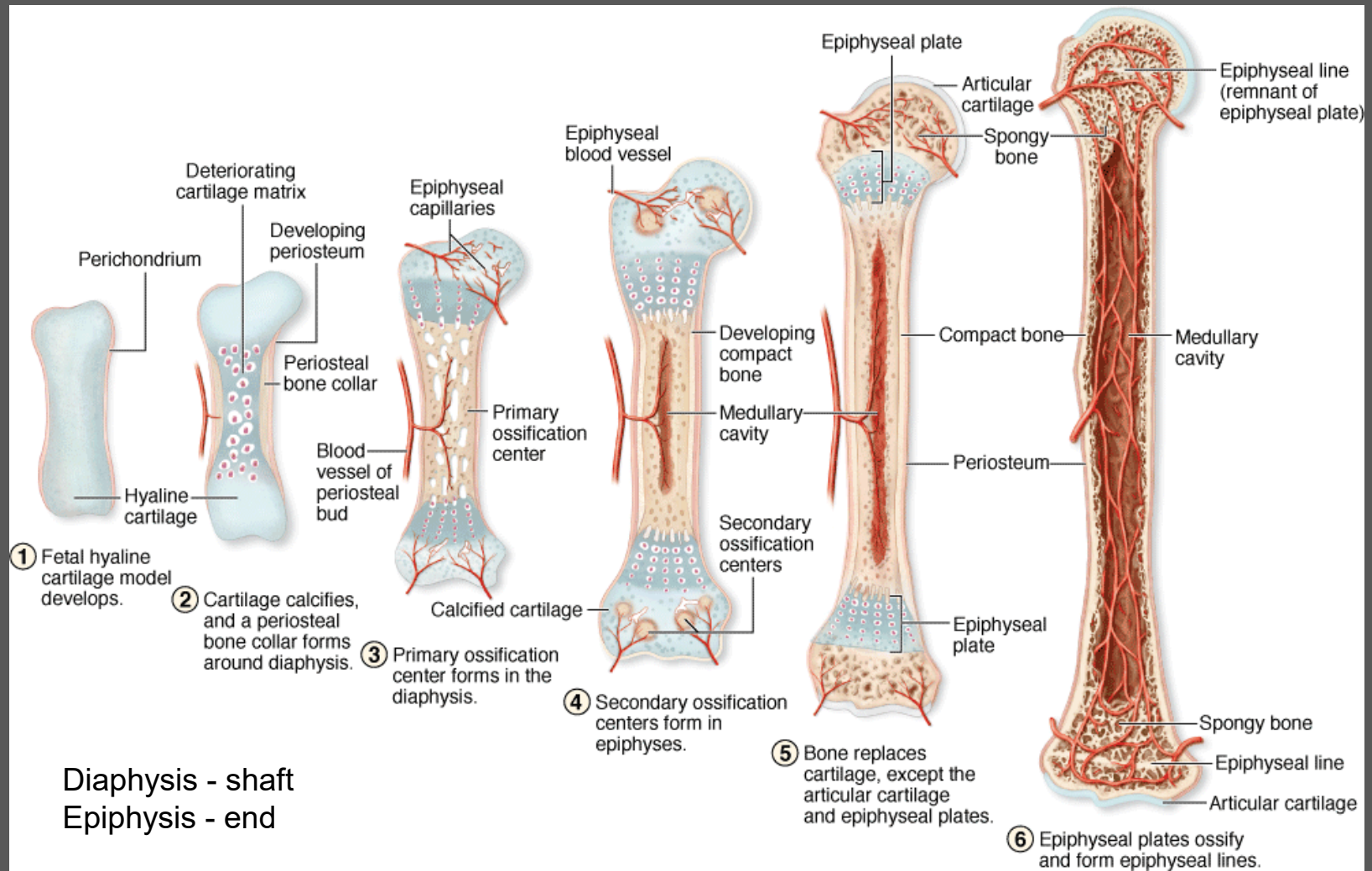
Blood Supply

- Nutrient foramina: Transmit nutrient arteries to supply marrow, spongy bone, and deeper compact bone.
- Haversian Systems: Allow for small blood vessels
 - osteocytes
- Veins accompany arteries; lymphatic vessels present



d Bryson: Cladonia Reso

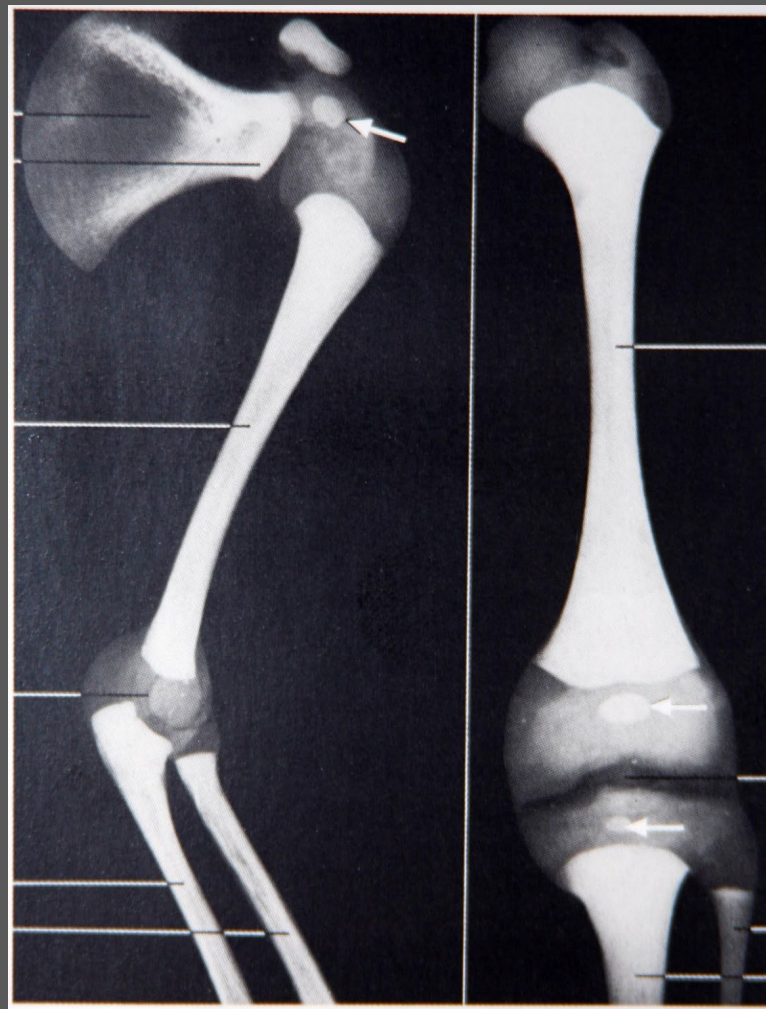
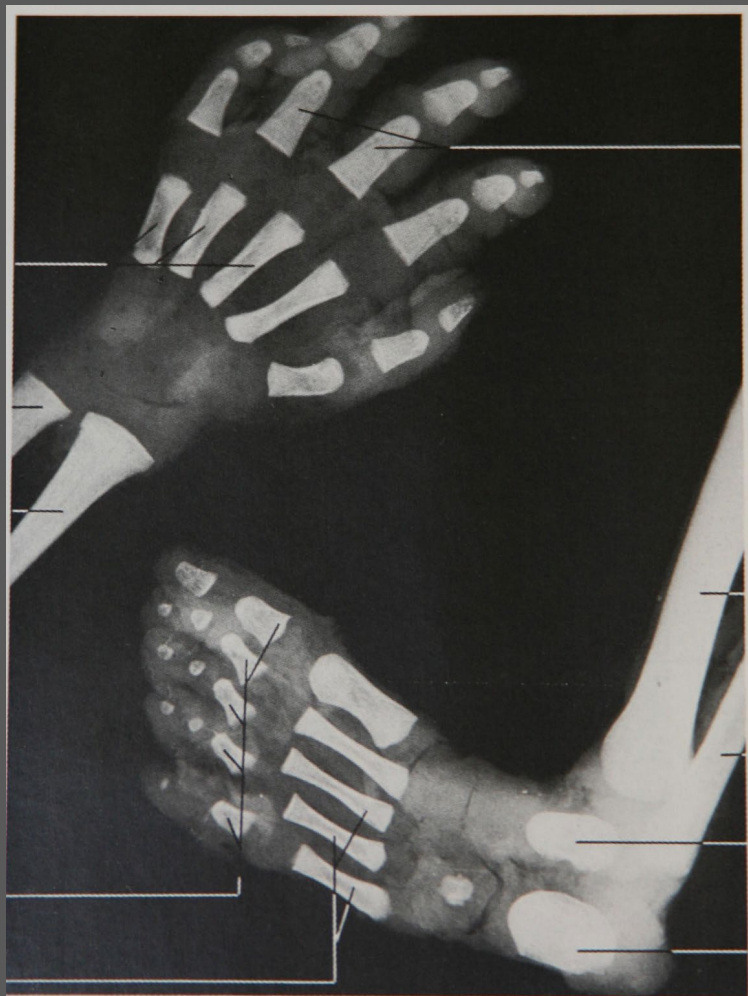
Endochondral Ossification



Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas*, 12th Edition: <http://www.accessmedicine.com>

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Cartilage models replaced by bone



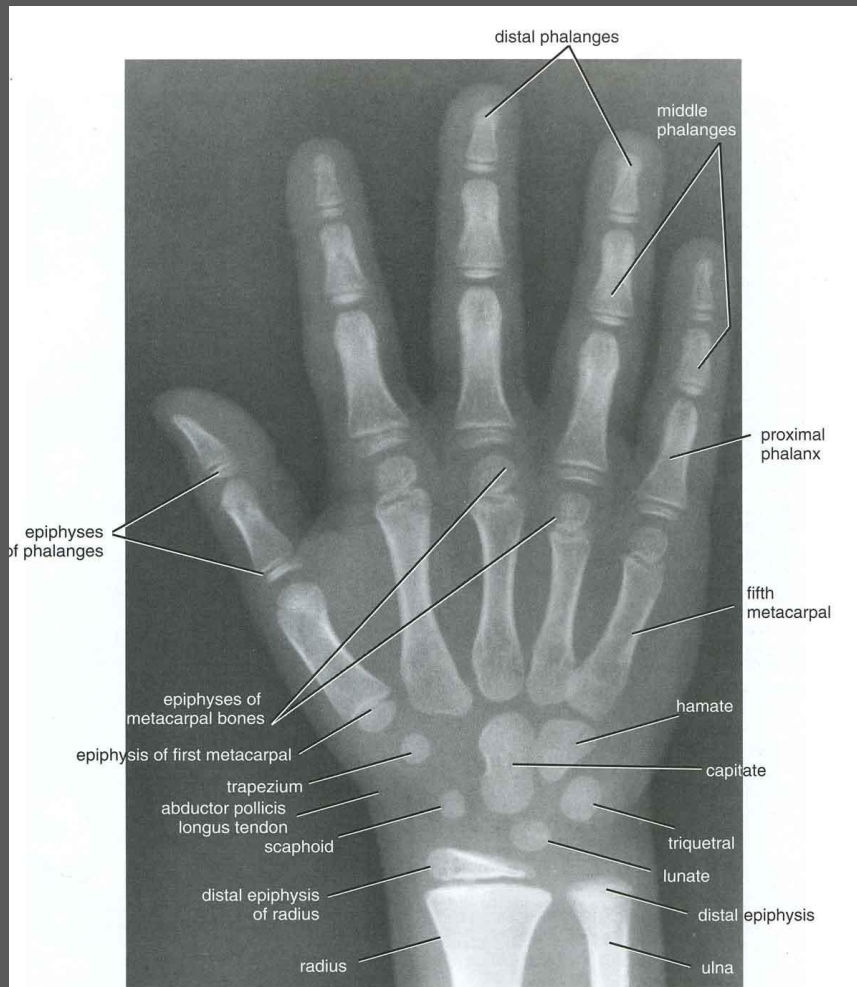


Figure 11-47 Posteroanterior radiograph of the wrist and hand of an 8-year-old boy.

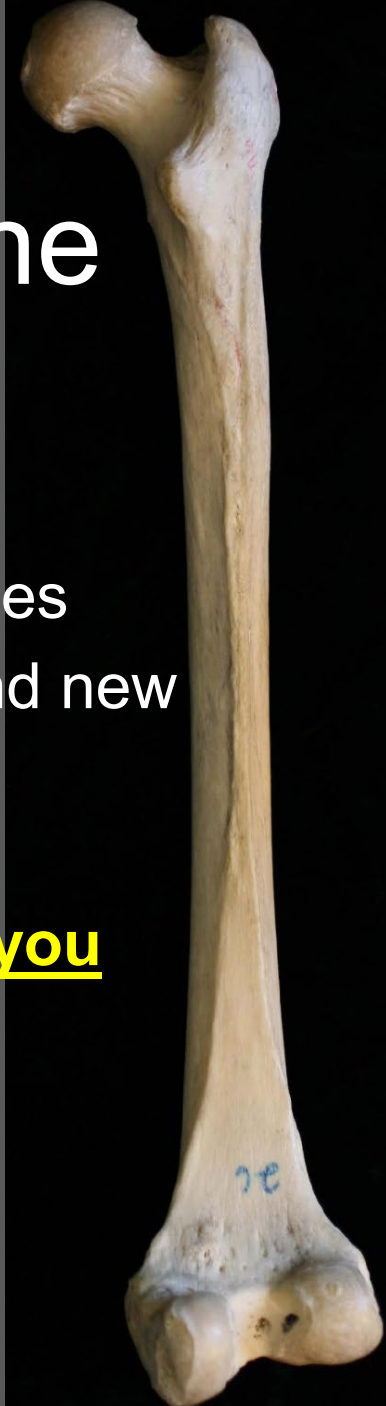
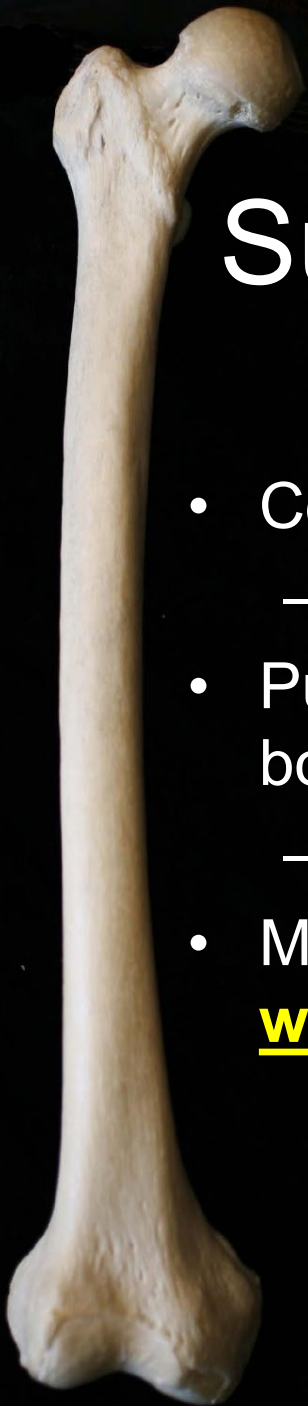
8 years



Adult

Surface markings of bone

- Connections for fibrous structures
 - fasciae, ligaments, tendons, aponeuroses
- Pull of fibrous tissue raises periosteum, and new bone deposited beneath
 - Become more obvious throughout life.
- Many markings have specific names **that you will need to know**



Introduction to Musculoskeletal Anatomy: Part 2- Cartilage and Joints

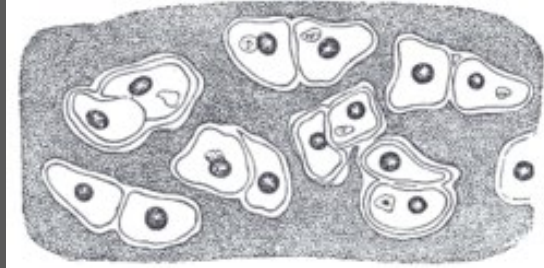
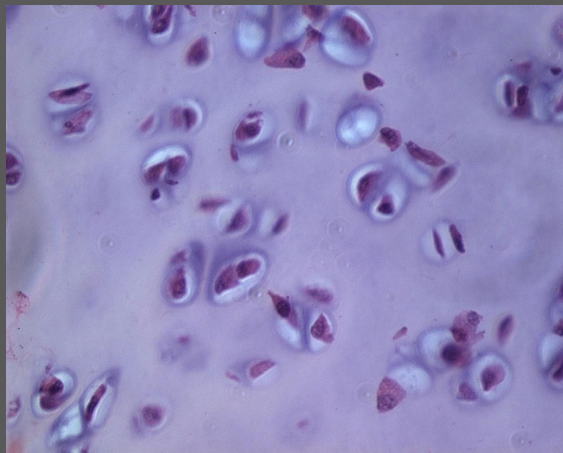
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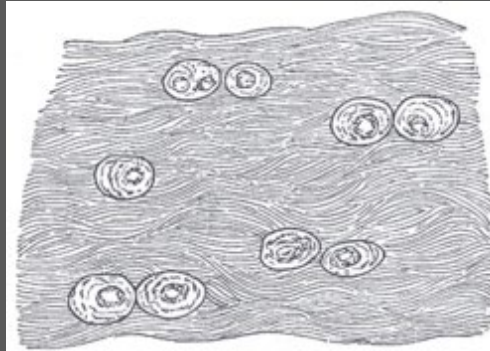


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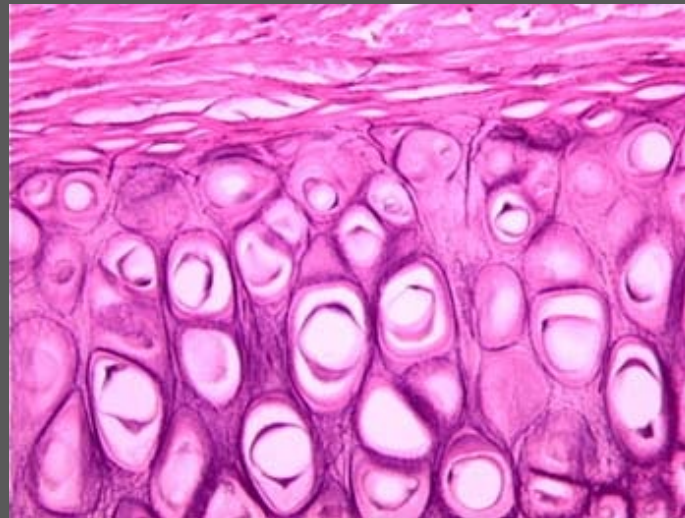
- Cartilage
 - Semirigid
 - Forms
 - where flexibility or padding is needed
 - articular surfaces of synovial joints
 - Avascular - obtains oxygen and nutrients by diffusion
 - Cartilage is covered by fibrous membrane (perichondrium) except on exposed joint surfaces.
 - Proportion of cartilage to bone changes with age



Hyaline cartilage

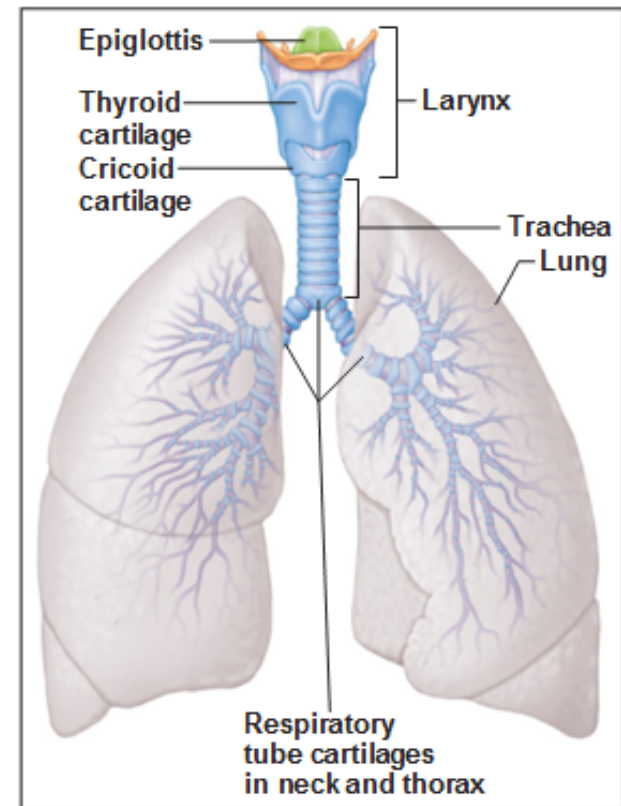
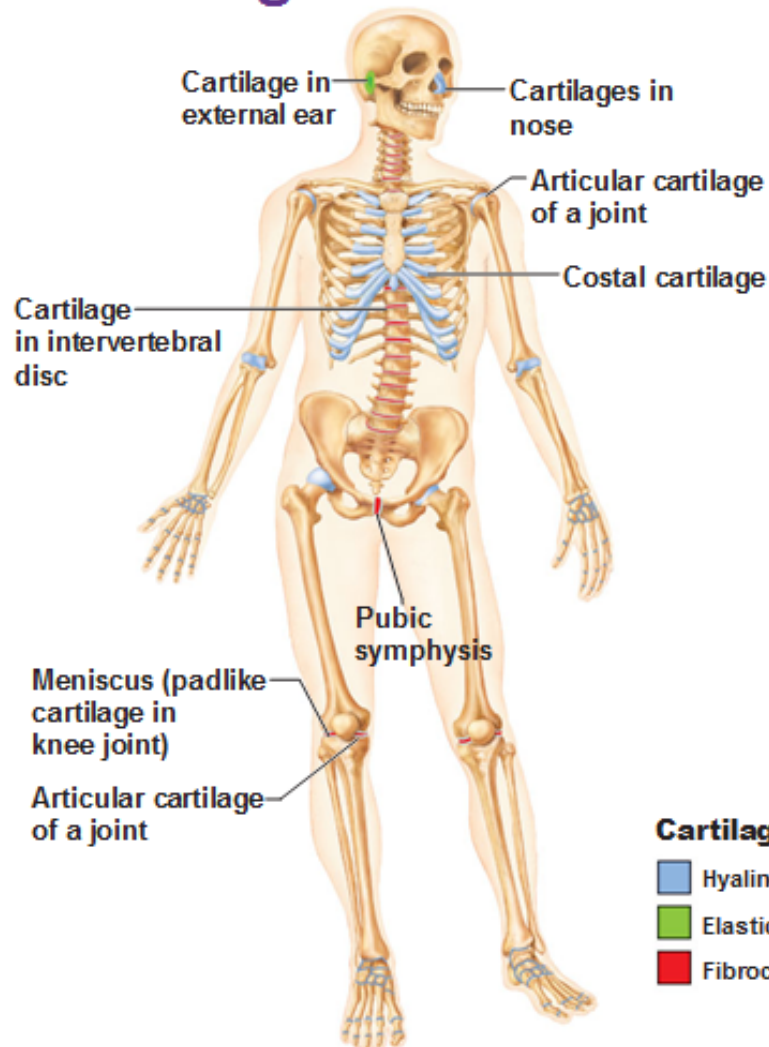


Fibrocartilage



Elastic cartilage

Cartilage - Locations



Cartilages

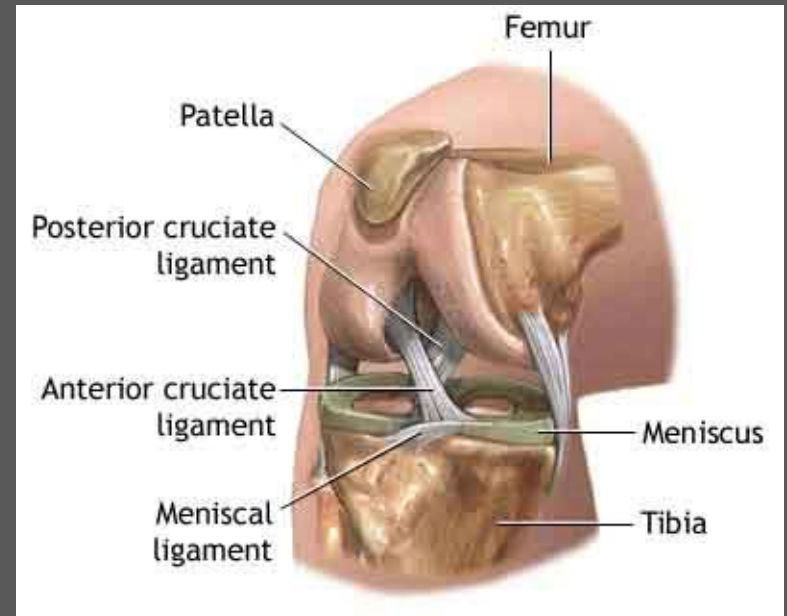
- Hyaline cartilages
- Elastic cartilages
- Fibrocartilages

Three types of cartilage

- Hyaline cartilage
 - long bones, epiphyseal plates, articular surfaces of synovial joints
- Fibrocartilage
 - discs within joints (e.g. menisci) of knee, glenoid labrum of shoulder
- Elastic cartilage
 - highly flexible (e.g. auricle of ear, epiglottis)

Joints

- Strength, stability, and motion of joint
 - Shapes of articular surfaces
 - Ligaments - prevent excessive movement
 - Muscles (muscle tone) e.g. rotator cuff
- Joint vasculature
 - Anastomoses assure blood supply to and across joints in various positions

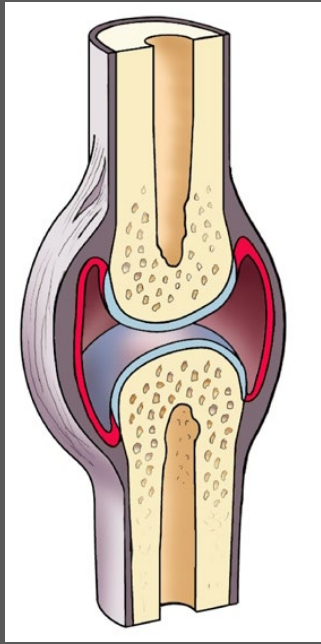


- Joint innervation
 - Hilton's law
 - Proprioception

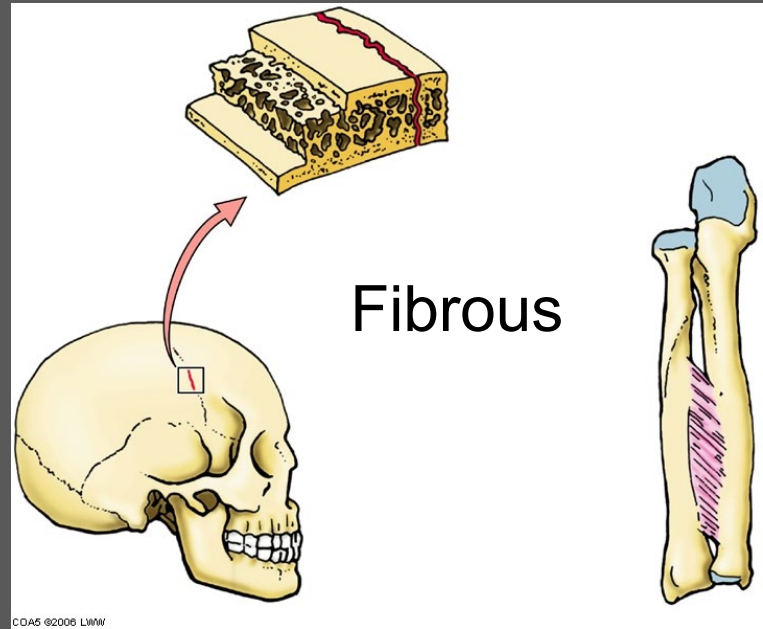
Hilton's law - the nerve supplying the muscles extending directly across and acting at a given joint not only supplies the muscle, but also innervates the joint and the skin overlying the muscle.



John Hilton 1805-1878



Synovial

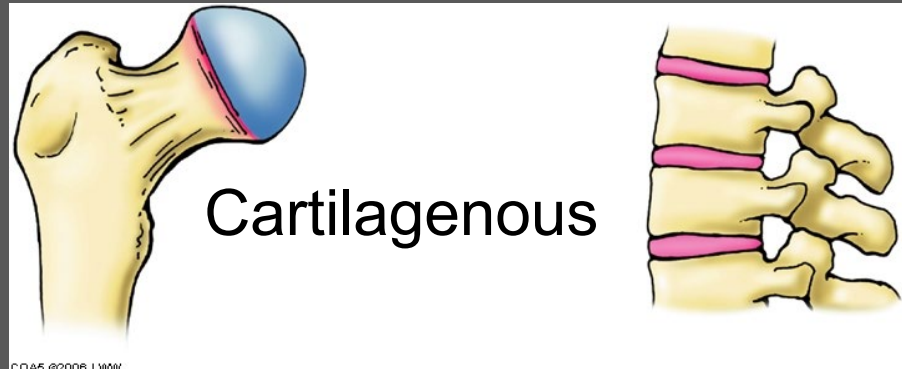


Fibrous

suture

syndesmosis

Types of joints

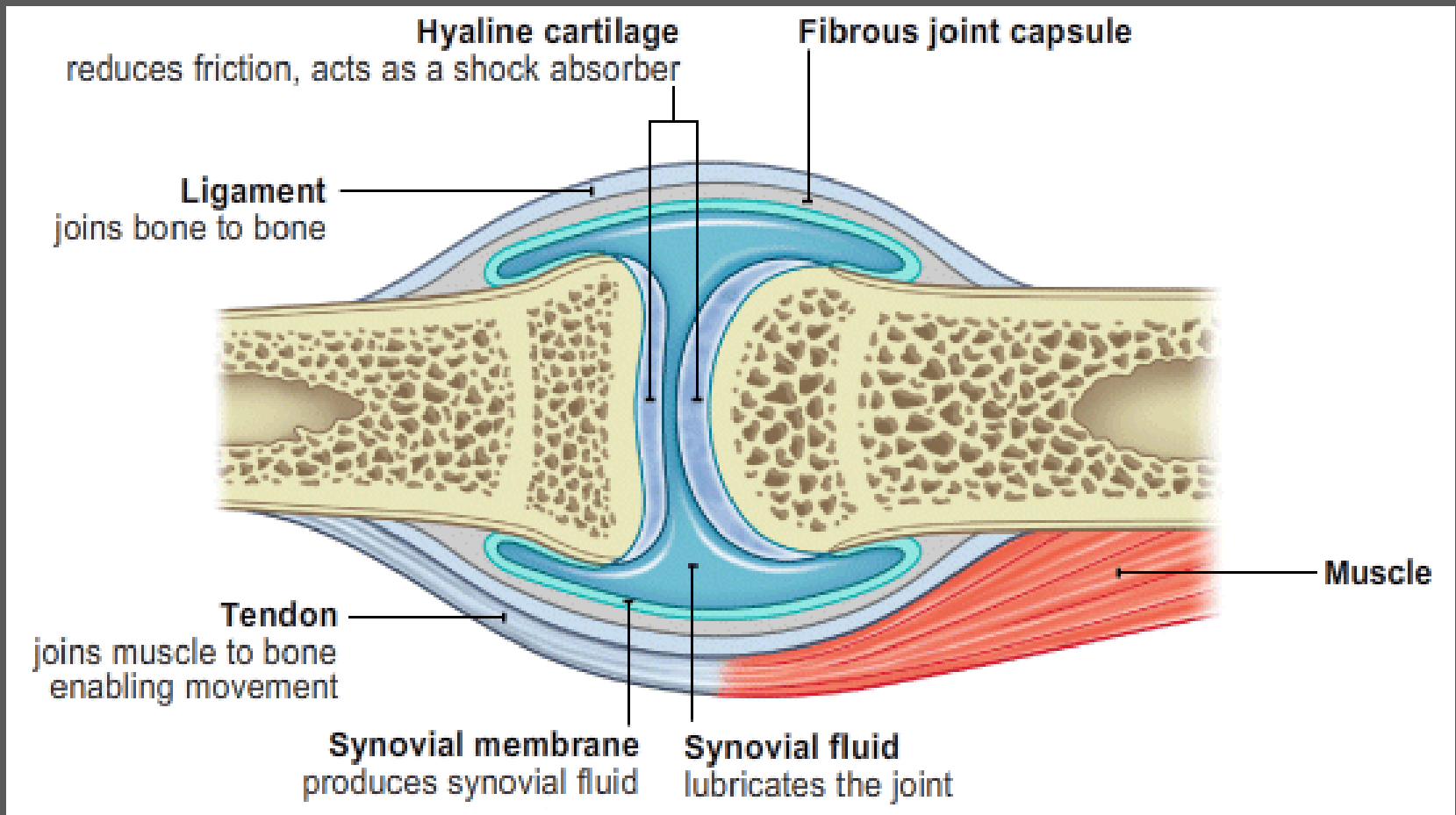


Cartilagenous

primary (synchondroses)

secondary

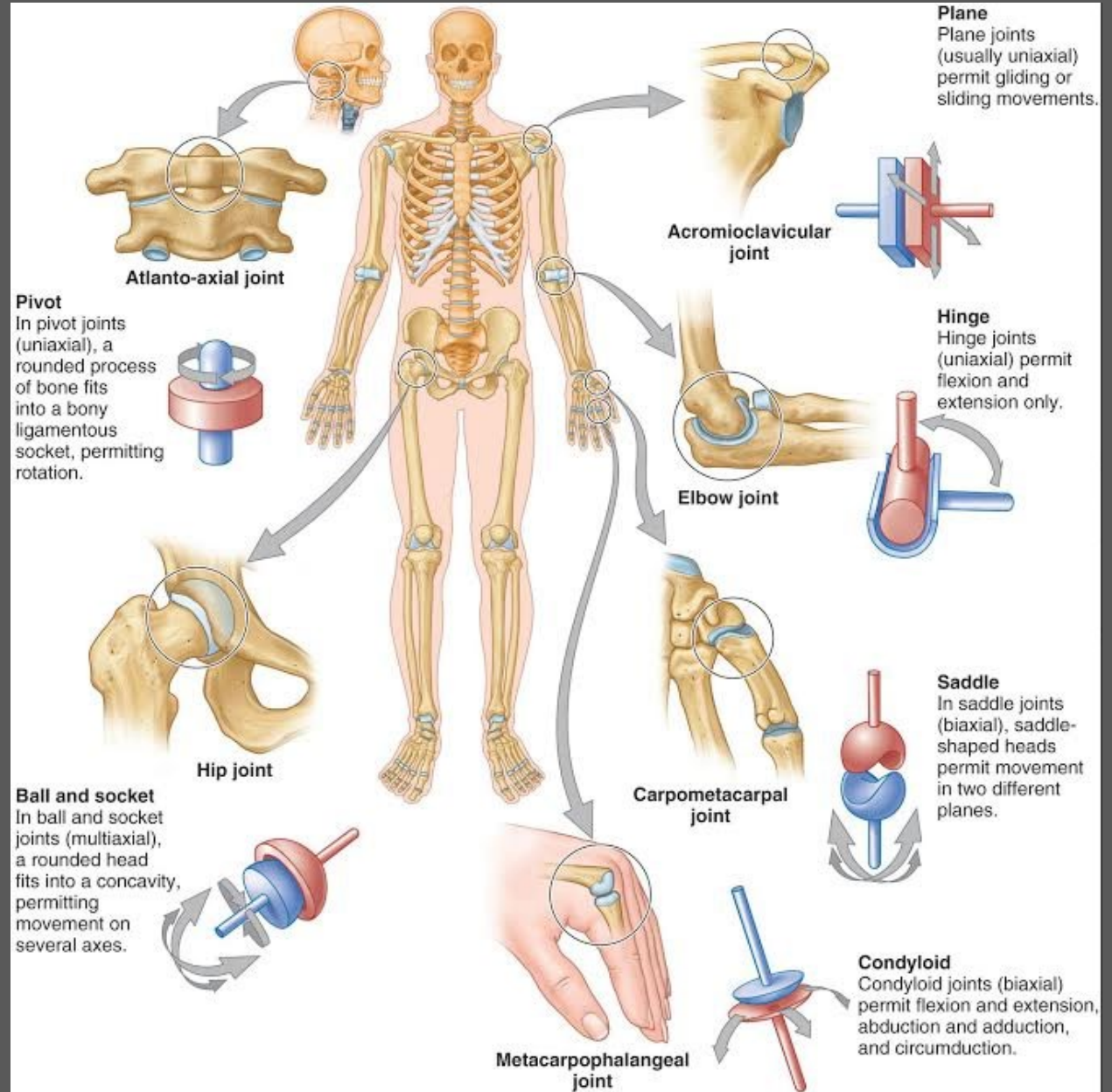
Synovial Joint





Major Types of Synovial Joints

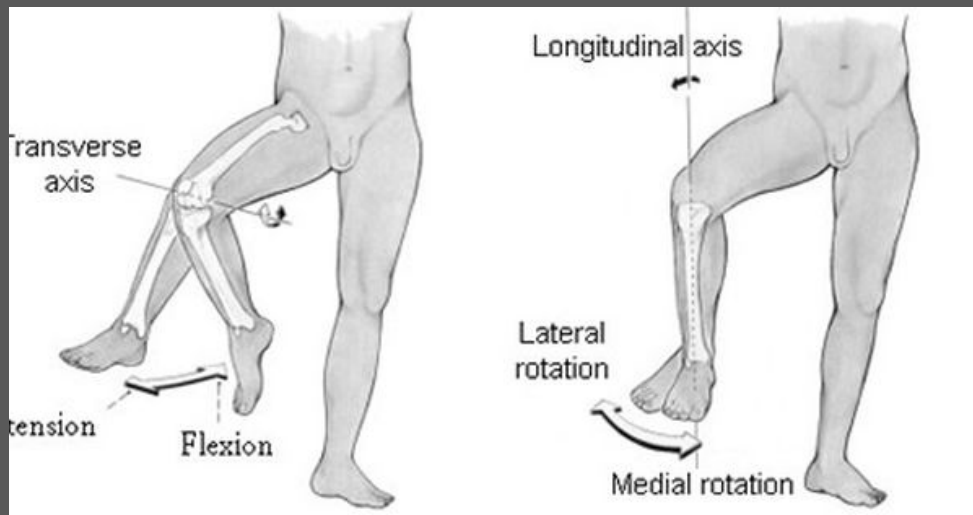
- Plane
- Pivot
- Hinge
- Saddle
- Condylloid
- Ball and socket
- Troch-
ginglymus
 - Flex, extend, rotate
 - knee



Which of the following motions are the metacarpophalangeal joints capable of?



What motions is the knee capable of?



Introduction to Musculoskeletal Anatomy: Part 3- Muscles and muscle Contractions

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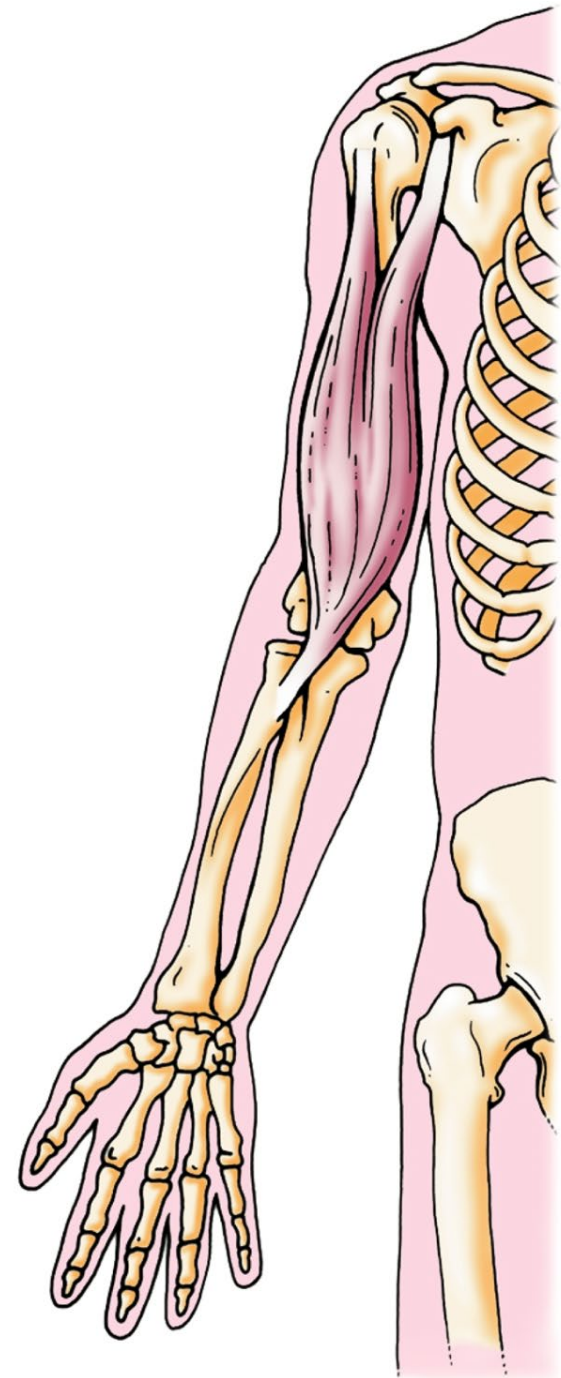
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Skeletal muscle

- Voluntary somatic muscle that moves or stabilizes structures
- Attached to bone, cartilage, ligaments, fascias, organs, skin
- Length: Belly + Tendon
- Named for action, position, length, or attachments.
 - Examples:
 - Adductor pollicis
 - Tibialis anterior



contraction

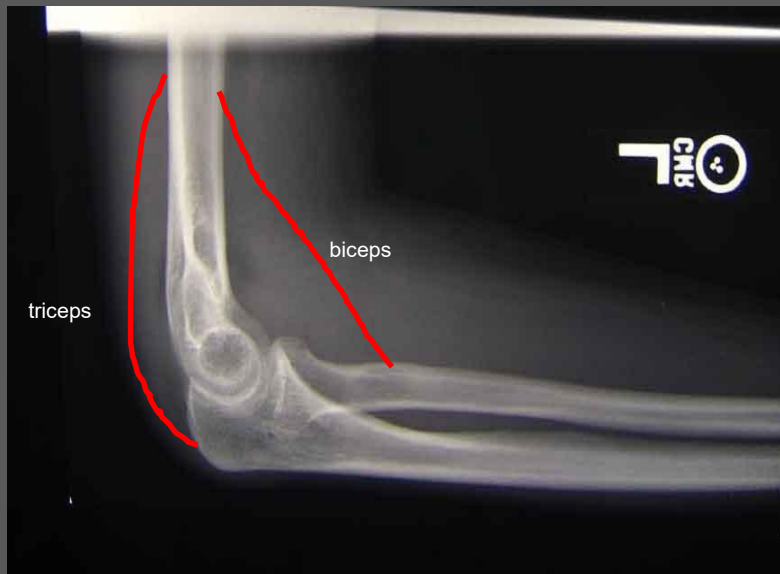
1.  the act of decreasing (something) in size or volume or quantity or scope
2.  (physiology) ~~a shortening or~~ tensing of a part or organ (especially of a muscle or muscle fiber)

A muscle contraction is the activation of muscle fibers resulting in the generation of tension.

A length change (in either direction) may occur but is not required.

Muscle Action

- The action of a muscle is what it does when it contracts and shortens with no resistance.
- In real life muscles have external resistance.

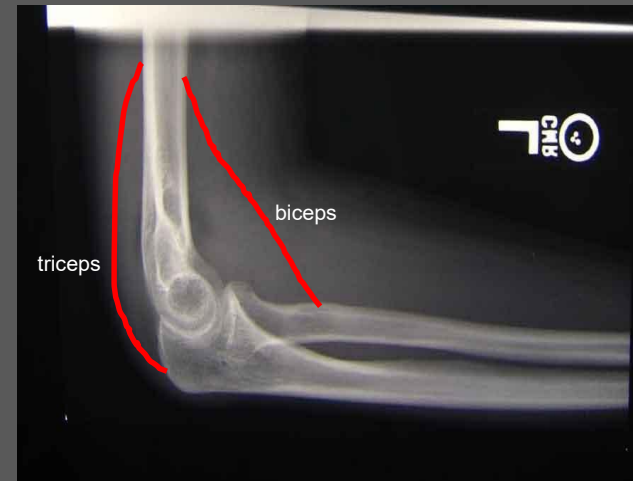


- Skeletal muscles function by contracting
 - Tonic – muscle tone
 - Isometric - muscle length does not change
 - Isotonic- muscle length changes, creates movement
 - concentric
 - eccentric

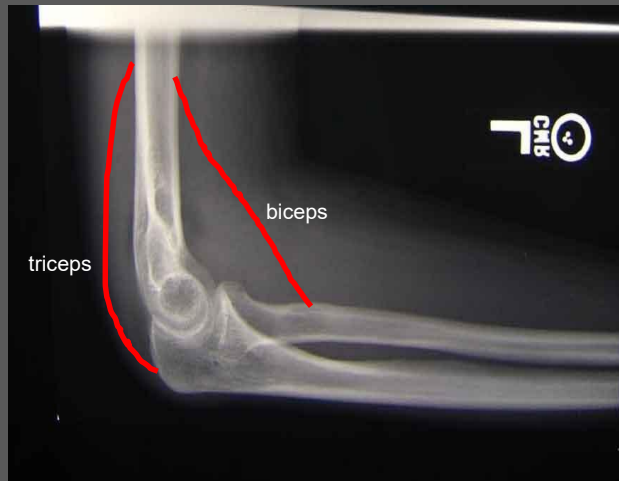


This scenario involves both flexion and extension at elbow.

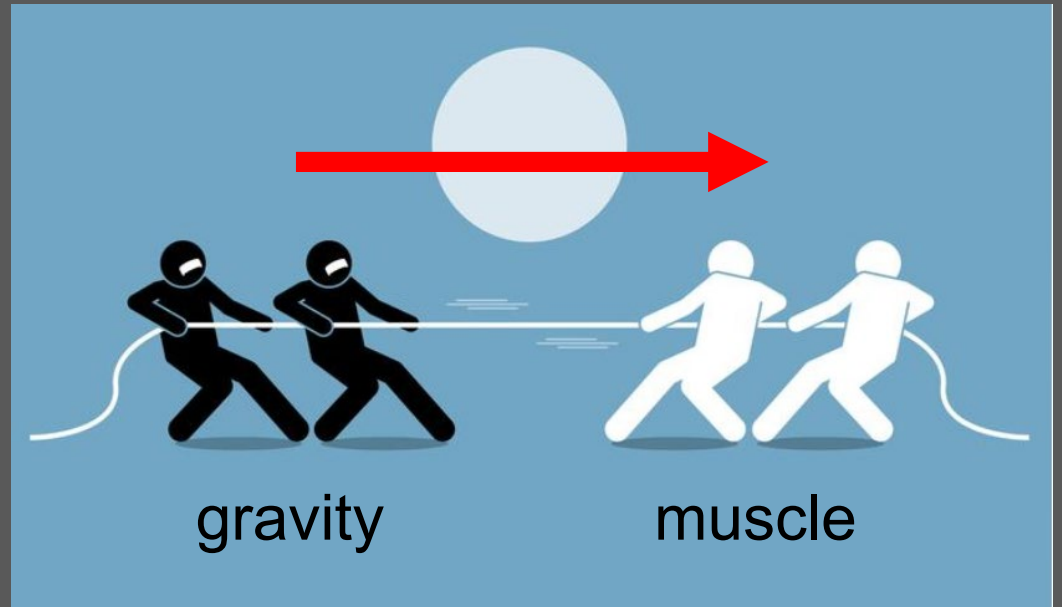
How does the biceps brachii control both of these actions, while the triceps brachii has no role?



- Isotonic contractions
 - Concentric - muscle shortens
 - Biceps brachii flexes elbow, lifts weight, overpowers gravity
 - Eccentric - muscle elongates
 - Weight (gravity) forces elbow to extend
 - Biceps brachii resists gravity, allows elbow to extend at controlled rate

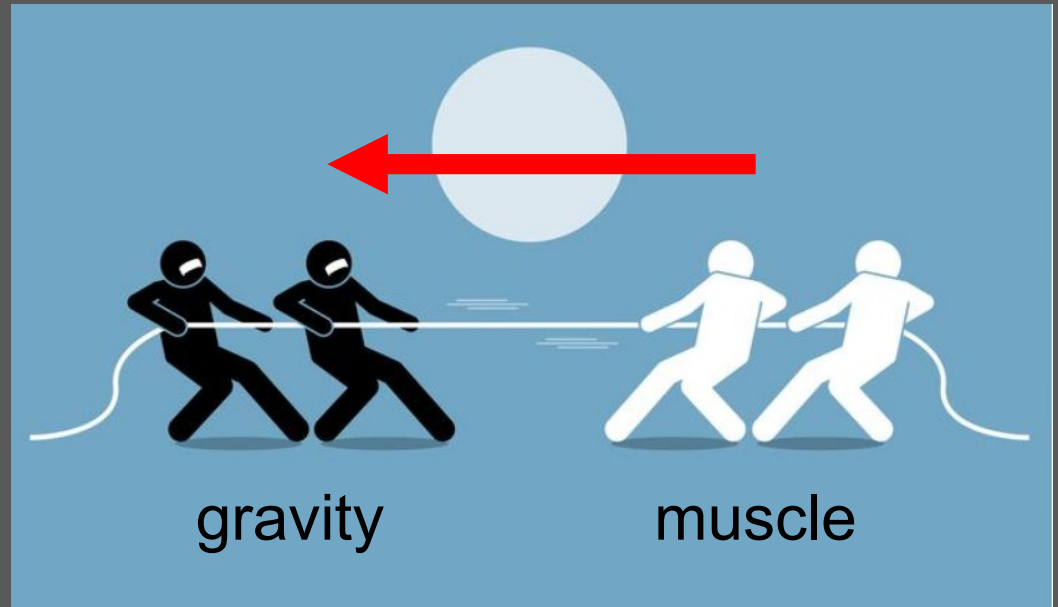


Concentric contraction of biceps brachii



Muscle overpowers gravity
Muscle shortens
Elbow flexes

Eccentric contraction of biceps brachii



Gravity overpowers muscle

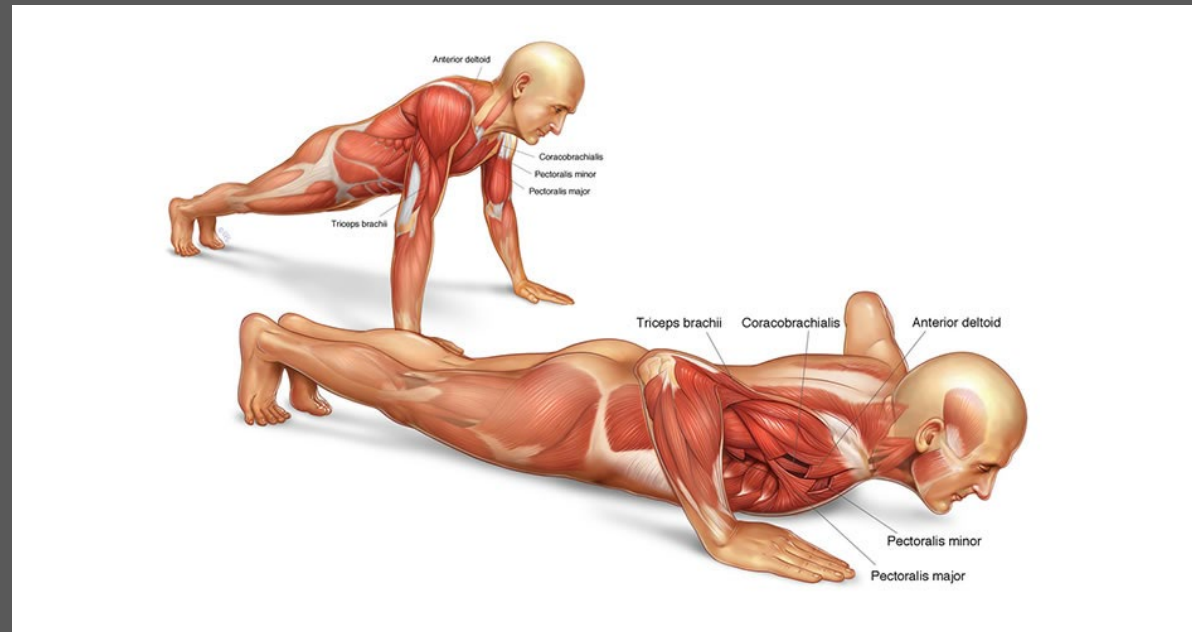
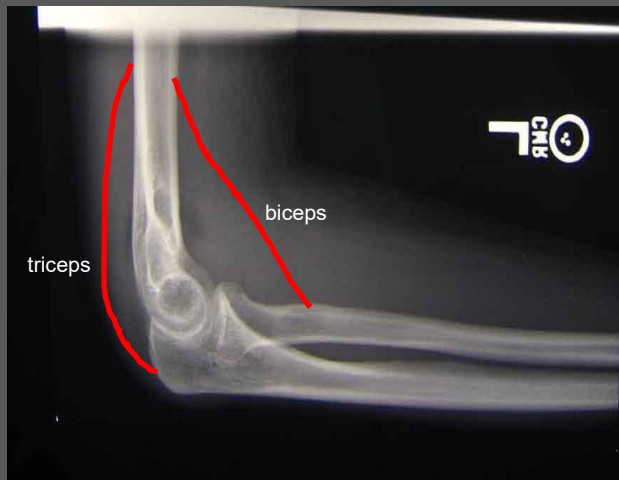
Muscle resists, but is elongated by gravity

Elbow extends at a slow, controlled rate

- What exercise could one do that involves both flexion and extension of the elbow via contractions of the triceps brachii?

Push-up

- What type of contraction is occurring in the triceps brachii doing during each phase of a push up?
 - Up phase
 - Down phase



Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>